DEPARTMENT OF INFORMATION TECHNOLOGY

SMT. PARMESHWARIDEVI DURGADUTT TIBREWALA

LIONS JUHU COLLEGE

OF ARTS, COMMERE AND SCIENCE

Affiliated to University of Mumbai

(Autonomous)

J.B. NAGAR, ANDHERI (E), MUMBAI-400059

Academic Year 2025-2026

SOFT COMPUTING

For

Semester I

Submitted By:

Mr. Ahmed Dilshad Shaikh

Msc.IT (Sem I)

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Certificate of Approval

This is to certify that practical entitled **"SOFT COMPUTING"**, Undertaken at **SMT.PARMESHWARIDEVI DURGADUTT TIBREWALA LIONS JUHU COLLEGE OF ARTS, COMMERECE & SCIENCE.** By **Mr. Ahmed Dilshad Shaikh** **Seat No.** **MIT251009** in partial fulfilment of **M.Sc. (IT) master degree (Semester I)** Examination had not been submitted for any other examination and does not form of any other course undergone by the candidate. It is further certified that he has completed all required phases of the practical.

Internal Examiner **External Examiner**

HOD / In-Charge / Coordinator **Signature/**

Principal/Stamp

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Practical 1a

Aim: Design a simple linear neural network model.

Code :

```
x=float(input("Enter value of x:"))
w=float(input("Enter value of weight w:"))
b=float(input("Enter value of bias b:"))
net = int(w*x+b)
if(net<0):
 out=0
elif((net>=0)&(net<=1)):
out =net
else:
 out=1
print("net=",net)
print("output=",out)
```

Output :

```
======== RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 1A.py ========
Enter value of x:3
Enter value of weight w:0.3
Enter value of bias b:0.5
net= 1
output= 1
```

Practical 1b

Aim: Calculate the output of neural net using both binary and bipolar sigmoidal function.

Code :

```
# number of elements as input
n = int(input("Enter number of elements : "))
# In[2]:
print("Enter the inputs")
inputs = [] # creating an empty list for inputs
# iterating till the range
for i in range(0, n):
 ele = float(input())
 inputs.append(ele) # adding the element
print(inputs)
# In[3]:
print("Enter the weights")
# creating an empty list for weights
weights = []
# iterating till the range
for i in range(0, n):
 ele = float(input())
 weights.append(ele) # adding the element
print(weights)
# In[4]:
print("The net input can be calculated as Yin = x1w1 + x2w2 + x3w3")
# In[5]:
Yin = []
for i in range(0, n):
Yin.append(inputs[i]*weights[i])
print(round(sum(Yin),3))
```

Output:

```
========= RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 1B.py =========
Enter number of elements : 3
Enter the inputs
0.5
0.6
0.2
[0.5, 0.6, 0.2]
Enter the weights
0.2
0.1
-0.3
[0.2, 0.1, -0.3]
The net input can be calculated as Yin = x1w1 + x2w2 + x3w3
0.1
```

Practical 2a

Aim: Generate AND/NOT function using McCulloch-Pitts neural net.

Code :

```
# enter the no of inputs
num_ip = int(input("Enter the number of inputs : "))
#Set the weights with value 1
w1 = 1
w2 = 1
print("For the ", num_ip , " inputs calculate the net input using yin = x1w1 + x2w2 ")
x1 = []
x2 = []
for j in range(0, num_ip):
 ele1 = int(input("x1 = "))
ele2 = int(input("x2 = "))
x1.append(ele1)
x2.append(ele2)
print("x1 = ",x1)
print("x2 = ",x2)
n = x1 * w1
m = x2 * w2
Yin = []
for i in range(0, num_ip):
 Yin.append(n[i] + m[i])
print("Yin = ",Yin)
#Assume one weight as excitatory and the other as inhibitory, i.e.,
Yin = []
for i in range(0, num_ip):
 Yin.append(n[i] - m[i])
print("After assuming one weight as excitatory and the other as inhibitory Yin = ",Yin)
#From the calculated net inputs, now it is possible to fire the neuron for input (1, 0)
#only by fixing a threshold of 1, i.e., θ ≥ 1 for Y unit.
#Thus, w1 = 1, w2 = -1; θ ≥ 1
Y=[]
for i in range(0, num_ip):
 if(Yin[i]>=1):
  ele= 1
Y.append(ele)
if(Yin[i]<1):
ele= 0
Y.append(ele)
print("Y = ",Y)
```

Output:

```
========= RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 2A.py =========
Enter the number of inputs : 4
For the  4  inputs calculate the net input using yin = x1w1 + x2w2
x1 = 0
x2 = 0
x1 = 0
x2 = 1
x1 = 1
x2 = 0
x1 = 1
x2 = 1
x1 =  [0, 0, 1, 1]
x2 =  [0, 1, 0, 1]
Yin =  [0, 1, 1, 2]
After assuming one weight as excitatory and the other as inhibitory Yin =  [0, -1, 1, 0]
Y =  [0, 0, 1, 0]
```

Practical 2b

Aim : Generate XOR function using McCulloch-Pitts neural net

Code:

```
import numpy as np
print('Enter weights')
w11=int(input('Weight w11='))
w12=int(input('weight w12='))
w21=int(input('Weight w21='))
w22=int(input('weight w22='))
v1=int(input('weight v1='))
v2=int(input('weight v2='))
print('Enter Threshold Value')
theta=int(input('theta='))
x1=np.array([0, 0, 1, 1])
x2=np.array([0, 1, 0, 1])
z=np.array([0, 1, 1, 0])
con=1
y1=np.zeros((4,))
y2=np.zeros((4,))
y=np.zeros((4,))
while con==1:
zin1=np.zeros((4,))
 zin2=np.zeros((4,))
 zin1=x1*w11+x2*w21
 zin2=x1*w21+x2*w22
 print("z1",zin1)
 print("z2",zin2)
 for i in range(0,4):
 if zin1[i]>=theta:
 y1[i]=1
  else:
   y1[i]=0
   if zin2[i]>=theta:
         y2[i]=1
   else:
    y2[i]=0
 yin=np.array([])
 yin=y1*v1+y2*v2
 for i in range(0,4):
 if yin[i]>=theta:
    y[i]=1
   else:
   y[i]=0
```

```
print("yin",yin)
print('Output of Net')
y=y.astype(int)
print("y",y)
print("z",z)
if np.array_equal(y,z):
 con=0
else:
 print("Net is not learning enter another set of weights and Threshold value")
 w11=input("Weight w11=")
 w12=input("weight w12=")
w21=input("Weight w21=")
w22=input("weight w22=")
v1=input("weight v1=")
v2=input("weight v2=")
theta=input("theta=")
print("McCulloch-Pitts Net for XOR function")
print("Weights of Neuron Z1")
print(w11)
print(w21)
print("weights of Neuron Z2")
print(w12)
print(w22)
print("weights of Neuron Y")
print(v1)
print(v2)
print("Threshold value")
print(theta)
```

output:

```
-------- RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 2B.py --------
Enter weights
Weight w11=1
weight w12=-1
weight w21=-1
weight w22=1
weight v1=1
weight v2=1
Enter Threshold Value
theta=1
z1 [ 0 -1  1  0]
z2 [ 0  1 -1  0]
yin [0. 1. 1. 0.]
Output of Net
y [0 1 1 0]
z [0 1 1 0]
theta=1
McCulloch-Pitts Net for XOR function
Weights of Neuron Z1
1
-1
weights of Neuron Z2
-1
1
weights of Neuron Y
1
1
Threshold value
1
```

Practical 3a

Aim : Write a program to implement Hebb's rule.

Code :

```
import numpy as np
#first pattern
x1=np.array([1,1,1,-1,1,-1,1,1,1])
#second pattern
x2=np.array([1,1,1,1,-1,1,1,1,1])
#initialize bais value
b=0
#define target
y=np.array([1,-1])
wtold=np.zeros((9,))
wtnew=np.zeros((9,))
wtnew=wtnew.astype(int)
wtold=wtold.astype(int)
bais=0
print("First input with target =1")
for i in range(0,9):
wtold[i]=wtold[i]+x1[i]*y[0]
wtnew=wtold
b=b+y[0]
print("new wt =", wtnew)
print("Bias value",b)
print("Second input with target =-1")
for i in range(0,9):
wtnew[i]=wtold[i]+x2[i]*y[1]
b=b+y[1]
print("new wt =", wtnew)
print("Bias value",b)
```

Output:

```
======== RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 3A.py ========
First input with target =1
new wt = [ 1  1  1 -1  1 -1  1  1  1]
Bias value 1
Second input with target =-1
new wt = [ 0  0  0 -2  2 -2  0  0  0]
Bias value 0
```

Practical 3b

Aim: Write a program to implement of delta rule.

Code :

```
#supervised learning
import numpy as np
import time
np.set_printoptions(precision=2)
x=np.zeros((3,))
weights=np.zeros((3,))
desired=np.zeros((3,))
actual=np.zeros((3,))
for i in range(0,3):
x[i]=float(input("Initial inputs:"))
for i in range(0,3):
 weights[i]=float(input("Initial weights:"))
for i in range(0,3):
desired[i]=float(input("Desired output:"))
a=float(input("Enter learning rate:"))
actual=x*weights
print("actual",actual)
print("desired",desired)
while True:
 if np.array_equal(desired,actual):
  break #no change
 else:
  for i in range(0,3):
 weights[i]=weights[i]+a*(desired[i]-actual[i])
 actual=x*weights
 print("weights",weights)
print("actual",actual)
print("desired",desired)
print("*"*30)
print("Final output")
print("Corrected weights",weights)
print("actual",actual)
print("desired",desired)
```

```
======== RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 3B.py ========
Initial inputs:1
Initial inputs:1
Initial inputs:1
Initial weights:1
Initial weights:1
Initial weights:1
Desired output:2
Desired output:3
Desired output:4
Enter learning rate:1
actual [1. 1. 1.]
desired [2. 3. 4.]
weights [2. 3. 4.]
actual [2. 3. 4.]
desired [2. 3. 4.]
******************************
Final output
Corrected weights [2. 3. 4.]
actual [2. 3. 4.]
desired [2. 3. 4.]
```

OUTPUT:

Practical 4a

Aim: Write a program for Back Propagation Algorithm

Code :

```
import numpy as np
import decimal
import math
np.set_printoptions(precision=2)
v1=np.array([0.6, 0.3])
v2=np.array([-0.1, 0.4])
w=np.array([-0.2,0.4,0.1])
b1=0.3
b2=0.5
x1=0
x2=1
alpha=0.25
print("calculate net input to z1 layer")
zin1=round(b1+ x1*v1[0]+x2*v2[0],4)
print("z1=",round(zin1,3))
print("calculate net input to z2 layer")
zin2=round(b2+ x1*v1[1]+x2*v2[1],4)
print("z2=",round(zin2,4))
print("Apply activation function to calculate output")
z1=1/(1+math.exp(-zin1))
z1=round(z1,4)
z2=1/(1+math.exp(-zin2))
z2=round(z2,4)
print("z1=",z1)
print("z2=",z2)
print("calculate net input to output layer")
yin=w[0]+z1*w[1]+z2*w[2]
print("yin=",yin)
print("calculate net output")
y=1/(1+math.exp(-yin))
print("y=",y)
fyin=y *(1- y)
dk=(1-y)*fyin
print("dk",dk)
dw1= alpha * dk * z1
dw2= alpha * dk * z2
dw0= alpha * dk
print("compute error portion in delta")
din1=dk* w[1]
din2=dk* w[2]
print("din1=",din1)
```

```
print("din2=",din2)
print("error in delta")
fzin1= z1 *(1-z1)
print("fzin1",fzin1)
d1=din1* fzin1
fzin2= z2 *(1-z2)
print("fzin2",fzin2)
d2=din2* fzin2
print("d1=",d1)
print("d2=",d2)
print("Changes in weights between input and hidden layer")
dv11=alpha * d1 * x1
print("dv11=",dv11)
dv21=alpha * d1 * x2
print("dv21=",dv21)
dv01=alpha * d1
print("dv01=",dv01)
dv12=alpha * d2 * x1
print("dv12=",dv12)
dv22=alpha * d2 * x2
print("dv22=",dv22)
dv02=alpha * d2
print("dv02=",dv02)
print("Final weights of network")
v1[0]=v1[0]+dv11
v1[1]=v1[1]+dv12
print("v=",v1)
v2[0]=v2[0]+dv21
v2[1]=v2[1]+dv22
print("v2",v2)
w[1]=w[1]+dw1
w[2]=w[2]+dw2
b1=b1+dv01
b2=b2+dv02
w[0]=w[0]+dw0
print("w=",w)
print("bias b1=",b1, " b2=",b2)
```

OUTPUT:

```
RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 4A.py
calculate net input to z1 layer
z1= 0.2
calculate net input to z2 layer
z2= 0.9
Apply activation function to calculate output
z1= 0.5498
z2= 0.7109
calculate net input to output layer
yin= 0.09101
calculate net output
y= 0.5227368084248941
dk 0.11906907074145694
compute error portion in delta
din1= 0.04762762829658278
din2= 0.011906907074145694
error in delta
fzin1 0.24751996
fzin2 0.20552119000000002
d1= 0.011788788650865037
d2 0.002447121711097841 7
Changes in weights between input and hidden layer
dv11= 0.0
dv21 0.002947197162716259 2
dv01= 0.002947197162716259 2
dv12= 0.0
dv22 0.000611780427774460 4
dv02= 0.000611780427774460 4
Final weights of network
v [0.6 0.3]
v2 [-0.1 0.4]
w= [ 0.17 0.42 0.12]
bias b1 0.3029471971627162 3 b2 0.5006117804277744
```

Practical 4b

Aim : Write a Program For Error Back Propagation Algorithm

Code :

```
import math
a0=-1
t=-1
w10=float(input("Enter weight first network"))
b10=float(input("Enter base first network:"))
w20=float(input("Enter weight second network:"))
b20=float(input("Enter base second network:"))
c=float(input("Enter learning coefficient:"))
n1=float(w10*c+b10)
a1=math.tanh(n1)
n2=float(w20*a1+b20)
a2=math.tanh(float(n2))
e=t-a2
s2=-2*(1-a2*a2)*e
s1=(1-a1*a1)*w20*s2
w21=w20-(c*s2*a1)
w11=w10-(c*s1*a0)
b21=b20-(c*s2)
b11=b10-(c*s1)
print("The updated weight of first n/w w11=",w11)
print("The uploaded weight of second n/w w21= ",w21)
print("The updated base of first n/w b10=",b10)
print("The updated base of second n/w b20= ",b20)
```

Output:

```
======== RESTART: C:\Users\chaud\Desktop\SCT Practicals\Practical 4B.py ========
Enter weight first network 12
Enter base first network:35
Enter weight second network:23
Enter base second network:45
Enter learning coefficient:11
The updated weight of first n/w w11= 12.0
The uploaded weight of second n/w w21=  23.0
The updated base of first n/w b10= 35.0
The updated base of second n/w b20=  45.0
```

Practical 5

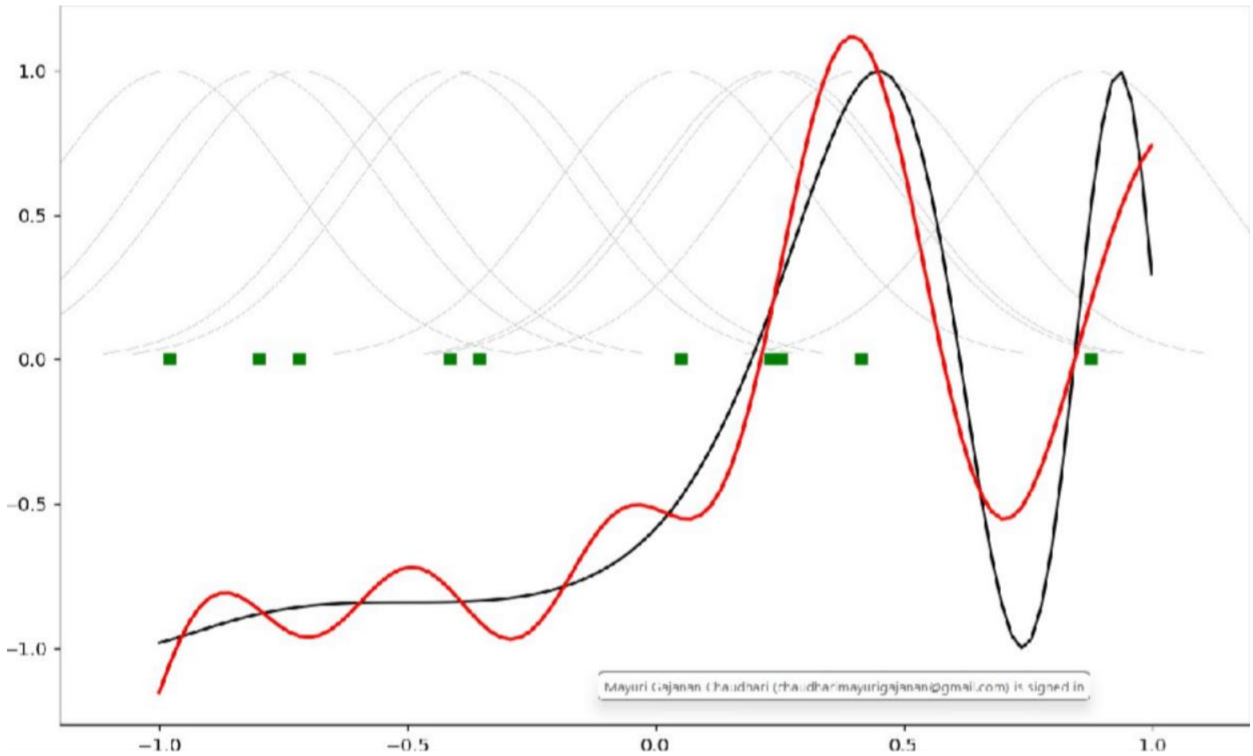
Aim: Write a program for Radial Basis function

Code:

```
from numpy import *
from scipy import *
from scipy.linalg import norm, pinv
from matplotlib import pyplot as plt
class RBF:
    def _init_(self, indim, numCenters, outdim):
        self.indim =indim
        self.outdim =outdim
        self.numCenters =numCenters
        self.centers =[random.uniform(-1, 1, indim) for i in range(numCenters)]
        self.beta =8
        self.W =random.random((self.numCenters, self.outdim))
    def _basisfunc(self, c, d):
        assert len(d) ==self.indim
        return exp(-self.beta *norm(c-d)**2)
    def _calcAct(self, X):
        G =zeros((X.shape[0], self.numCenters), float)
        for ci, c in enumerate(self.centers):
            for xi, x in enumerate(X):
                G[xi,ci] =self._basisfunc(c, x)
        return G
    def train(self, X, Y):
        """ X: matrix of dimensions n x indim
         y: column vector of dimension n x 1 """
        # choose random center vectors from training set
        rnd_idx =random.permutation(X.shape[0])[:self.numCenters]
        self.centers =[X[i,:] for i in rnd_idx]
        print("center", self.centers)
        # calculate activations of RBFs
        G =self._calcAct(X)
        print (G)
        # calculate output weights (pseudoinverse)
        self.W =dot(pinv(G), Y)
    def test(self, X):
        """ X: matrix"""
        G =self._calcAct(X)
        Y =dot(G, self.W)
        return Y
if _name=='_main_':  #
      ------ 1D Example ----------------------------------------------------
    n =100
```

```
x =mgrid[-1:1:complex(0,n)].reshape(n, 1)
# set y and add random noise
y =sin(3*(x+0.5)**3-1)
    # y += random.normal(0, 0.1, y.shape)
# rbf regression

rbf =RBF(1, 10, 1)
rbf.train(x, y)
z =rbf.test(x)
# plot original data
plt.figure(figsize=(12, 8))
plt.plot(x, y, 'k-')
# plot learned model
plt.plot(x, z, 'r-', linewidth=2)
# plot rbfs
plt.plot(rbf.centers, zeros(rbf.numCenters), 'gs')
for c in rbf.centers:
    # RF prediction lines
    cx =arange(c-0.7, c+0.7, 0.01)
    cy =[rbf._basisfunc(array([cx_]), array([c])) for cx_ in cx]
    plt.plot(cx, cy, '-', color='gray', linewidth=0.2)
plt.xlim(-1.2, 1.2)
plt.show()
```

Output:



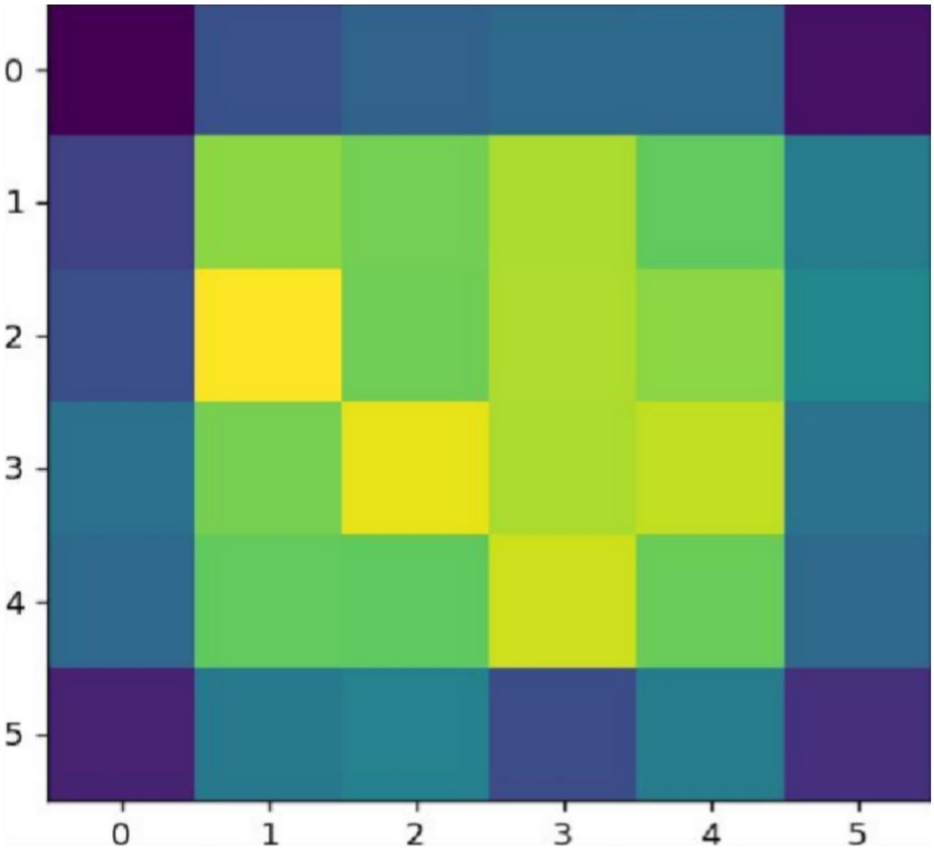
Practical 6

Aim: Write a program for Kohonen Self organizing map

Code :

```
from minisom import MiniSom
import matplotlib.pyplot as plt
data = [[ 0.80, 0.55, 0.22, 0.03],
[ 0.82, 0.50, 0.23, 0.03],
[ 0.80, 0.54, 0.22, 0.03],
[ 0.80, 0.53, 0.26, 0.03],
[ 0.79, 0.56, 0.22, 0.03],
[ 0.75, 0.60, 0.25, 0.03],
[ 0.77, 0.59, 0.22, 0.03]]
som = MiniSom(6, 6, 4, sigma=0.3, learning_rate=0.5) # initialization of 6x6 SOM
som.train_random(data, 100) # trains the SOM with 100 iterations
plt.imshow(som.distance_map())
plt.show()
```

Output:



Practical 7

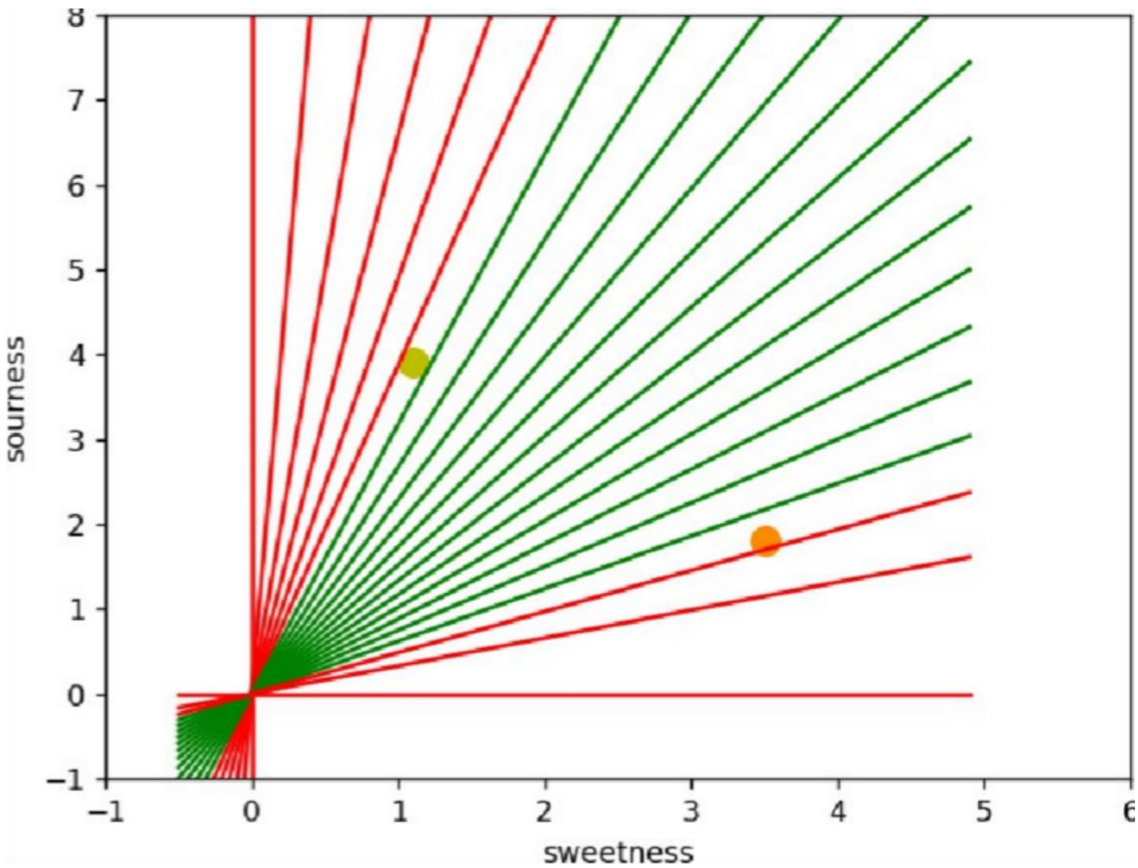
Aim: Write a program for Linear separation.

Code :

```
import numpy as np
import matplotlib.pyplot as plt
def create_distance_function(a, b, c):
 """ 0 = ax + by + c """
 def distance(x, y):
  """ returns tuple (d, pos)
   d is the distance
   If pos == -1 point is below the line,
  0 on the line and +1 if above the line
  """
  nom = a * x + b * y + c
  if nom == 0:
   pos = 0
  elif (nom<0 and b<0) or (nom>0 and b>0):
   pos = -1
  else:
   pos = 1
  return (np.absolute(nom) / np.sqrt( a ** 2 + b ** 2), pos)
 return distance
points = [ (3.5, 1.8), (1.1, 3.9) ]
fig, ax = plt.subplots()
ax.set_xlabel("sweetness")
ax.set_ylabel("sourness")
ax.set_xlim([-1, 6])
ax.set_ylim([-1, 8])
X = np.arange(-0.5, 5, 0.1)
colors = ["r", ""] # for the samples
size = 10
for (index, (x, y)) in enumerate(points):
 if index== 0:
  ax.plot(x, y, "o", color="darkorange", markersize=size)
 else:
  ax.plot(x, y, "oy", markersize=size)
  step = 0.05
for x in np.arange(0, 1+step, step):
 slope = np.tan(np.arccos(x))
 dist4line1 = create_distance_function(slope, -1, 0)
 #print("x: ", x, "slope: ", slope)
 Y = slope * X
 results = []
 for point in points:
```

```
  results.append(dist4line1(*point))
  #print(slope, results)
 if (results[0][1] != results[1][1]):
  ax.plot(X, Y, "g-")
 else:
  ax.plot(X, Y, "r-")
plt.show()
```

Output:



Practical 8A

Aim: Membership and Identity Operators | in, not in,

Code :

```
# Python program to illustrate
# Finding common member in list
# without using 'in' operator
# Define a function() that takes two lists
def overlapping(list1,list2):
 c=0
 d=0
 for i in list1:
  c+=1
 for i in list2:
  d+=1
 for i in range(0,c):
  for j in range(0,d):
  if(list1[i]==list2[j]):
  return 1
 return 0
list1=[1,2,3,4,6]
list2=[6,7,8,9]
if(overlapping(list1,list2)):
 print("overlapping")
else:
 print("not overlapping")
```

Output: overlapping

Practical 8b

Aim: Membership and Identity Operators is, is not

Code :

```
# Python program to illustrate the use
# of 'is' identity operator
'''x = 5
if (type(x) is int):
 print ("true")
else:
 print ("false")'''
# Python program to illustrate the
# use of 'is not' identity operator
x = 1
if (type(x) is not int):
 print ("true")
else:
 print ("false")
```

Output: false

Practical 9a

Aim: Find the ratios using fuzzy logic

Code :

```
# Python code showing all the ratios together,
# make sure you have installed fuzzywuzzy module
from fuzzywuzzy import fuzz
from fuzzywuzzy import process
s1 = "I love fuzzysforfuzzys"
s2 = "I am loving fuzzysforfuzzys"
print ("FuzzyWuzzy Ratio:", fuzz.ratio(s1, s2))
print ("FuzzyWuzzyPartialRatio: ", fuzz.partial_ratio(s1, s2))
print ("FuzzyWuzzyTokenSortRatio: ", fuzz.token_sort_ratio(s1, s2))
print ("FuzzyWuzzyTokenSetRatio: ", fuzz.token_set_ratio(s1, s2))
print ("FuzzyWuzzyWRatio: ", fuzz.WRatio(s1, s2),'\n\n')
# for process library,
query = 'fuzzys for fuzzys'
choices = ['fuzzy for fuzzy', 'fuzzy fuzzy', 'g. for fuzzys']
print ("List of ratios: ")
print (process.extract(query, choices), '\n')
print ("Best among the above list: ",process.extractOne(query, choices))
```

Output:

```
FuzzyWuzzy Ratio: 86
FuzzyWuzzyPartialRatio:  86
FuzzyWuzzyTokenSortRatio:  86
FuzzyWuzzyTokenSetRatio:  87
FuzzyWuzzyWRatio:  86


List of ratios:
[('g. for fuzzys', 95), ('fuzzy for fuzzy', 94), ('fuzzy fuzzy', 86)]

Best among the above list:  ('g. for fuzzys', 95)
```

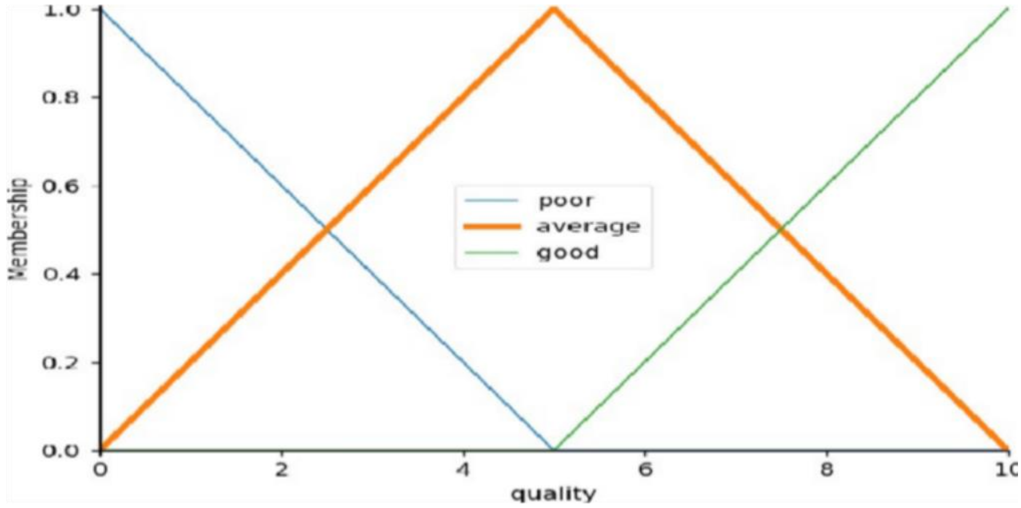
Practical 9b

Aim: Solve Tipping Problem using fuzzy logic

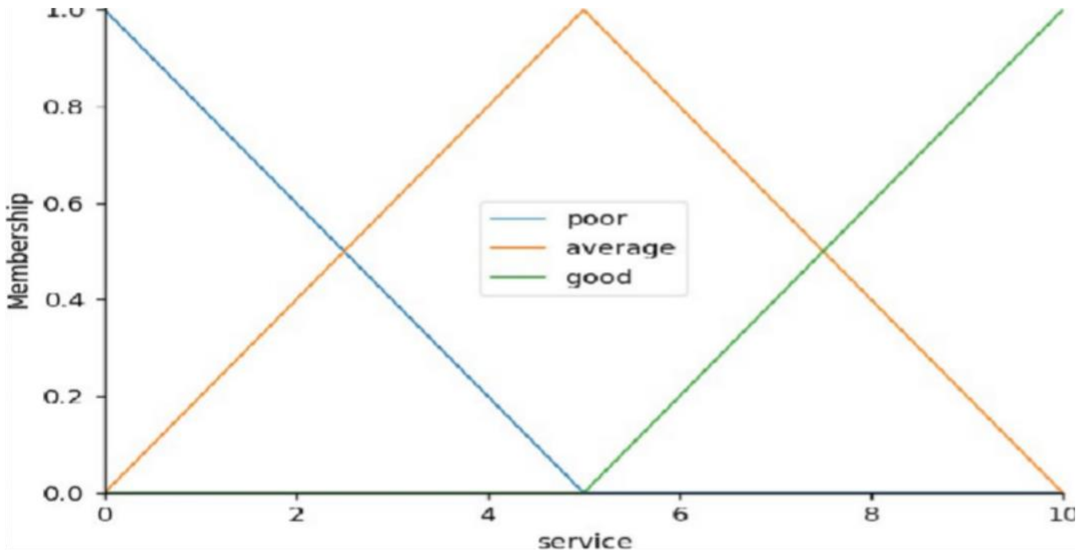
Code :

```
import numpy as np

import skfuzzy as fuzz

from skfuzzy import control as ctrl

quality = ctrl.Antecedent(np.arange(0, 11, 1), 'quality')

service = ctrl.Antecedent(np.arange(0, 11, 1), 'service')

tip = ctrl.Consequent(np.arange(0, 26, 1), 'tip')

quality.automf(3)

service.automf(3)

tip['low'] = fuzz.trimf(tip.universe, [0, 0, 13])

tip['medium'] = fuzz.trimf(tip.universe, [0, 13, 25])

tip['high'] = fuzz.trimf(tip.universe, [13, 25, 25])

quality['average'].view()

service.view()

tip.view()

rule1 = ctrl.Rule(quality['poor'] | service['poor'], tip['low'])

rule2 = ctrl.Rule(service['average'], tip['medium'])

rule3 = ctrl.Rule(service['good'] | quality['good'], tip['high'])

rule1.view()

tipping_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])

tipping = ctrl.ControlSystemSimulation(tipping_ctrl)

# Pass inputs to the ControlSystem using Antecedent labels with Pythonic API

tipping.input['quality'] = 6.5

tipping.input['service'] = 9.8

# Crunch the numbers

tipping.compute()

print(tipping.output['tip'])

tip.view(sim=tipping)
```

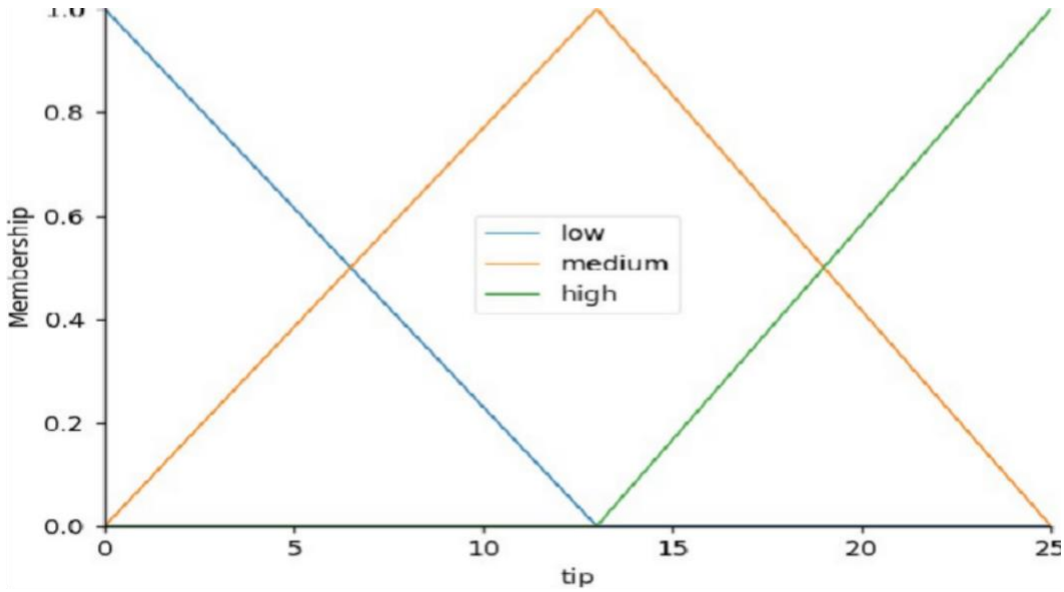
Output:



Membership
quality
poor
average
good



Membership
service
poor
average
good



Membership
tip
low
medium
high